

NX-3210R

VHF DIGITAL TRANSCEIVER

THE CLEAR CHOICE FOR RAIL OPERATORS ACROSS NORTH AMERICA

The NX-3210R portable two way radio, latest addition to the NEXEDGE® lineup, succeeds the proven NX-210G. Featuring a backlit LCD, large keypad, 7-color lightbar, Voice Guide, and built-in AAR feature, it offers superb operational ease and convenience. And with its built in GPS receiver, solid performance and MIL STD 810 qualifications, it is sure to attract customers throughout the North American rail market. To meet their needs, support is provided in the USA and Canada for customizing railroad specific features. The NX-3210R setting the benchmark for vital rail communications.

Features

TX power 5 / 4 / 1 W operation

Mixed Protocol Operation: NXDN Type-C & Gen2 Trunked, NXDN Conventional and FM analog only models are available to meet various demands

Mixed Protocol Zones: Each channel in a zone can be from a different system

512 CH-GID Optional 1000 CH-GID also available

GPS Data with built-in GPS receiver

1W speaker audio at 5% distortion

SDM Manual Input

Transparent Data Mode

Voice Guide/Voice & GPS Data storage with internal memory

RSSI information is included in GPS Data (Conventional mode)

AAR Functions and Channels.

LCD display: 4-line basic frame: 2-line main/ sub display, icon & key guides

14-character, 5-line text message frame: 3 lines of text text, icon & key guides

Display in 3 languages: English, Spanish, and French

12-/24-hr clock and date display

Keys: Front: Cross keys and 16 keypads / Side: 2 PF keys / Top: Emergency/AUX key

On/Off Volume knob

7-color Light Bar

PC serial interface

Built-in Bluetooth® 4.0

Feedback Suppression

Scan: Single Zone / Multi-Zone / List Scan

Dual Priority scan (conventional)

MIL-STD-810 C/D/E/F/G/H

IP54/55 Water & Dust intrusion

Analog Mode - General

12.5 & 25 kHz Channels

Conventional & LTR

FleetSync/II, MDC-1200, DTMF

QT/DQT/2-tone (Conventional)

Internal Voice Scrambler

NXDN™ **Gen2** **Bluetooth®** **FleetSync®** **GPS**



Digital Mode - General

NXDN Digital Air Interface

AMBE+2 Vocoder

6.25 & 12.5 kHz Channels

Over-the-Air Alias

OTAP (included in AAR mode)

Paging Call

Emergency Call

All Group Call

Status Messaging

Remote Stun/Kill

Short & Long Data Messages

GPS Data with Voice

NXDN Scrambler

DES & AES Encryption (w/ KPG -DE1/AE1 and w/o KWD-AE31)

Digital Mode - Conventional

64 Radio Access Numbers (RAN)

Individual & Group Selective Call

Mixed FM/Digital Operation

Conventional IP Networks

Site Roaming

Digital Mode - Trunking

Individual Private Call

Transmission Trunked Mode

Message Trunked Mode

Call Queuing With Priority

4 Priority Monitor IDs

Remote Group Add

Fail Soft Mode

Multi-Site IP Networks Compatible

60,000 GIDs / UIDs

Wide Area Group call

Auto Roaming Registration

Group Registration

Multi-System Compatible

24-trunked Networks

UID List for each network

Accessories

All accessories may not be available in all markets.
Contact an authorized KENWOOD dealer for details and complete list of all accessories.

KNB-L1/L2/L11 L12/L13 Li-ion Battery Pack



KSC-52BK Single Unit Rapid Charger (For KNB-L1/L2)



KSC-52AK Single Unit Rapid Charger (For KNB-L11 L12/L13)



KMB-526K Wall Mount Bracket for KSC-526K



KSC-526K Rapid 6-unit Charger Base



(W/ Pocket KSC-52PBW for KNB-L1/L2,
and W/ Pocket KSC-52PAW for KNB-L11/L12/L13)

KRA-22 VHF Helical Antenna (Low Profile)



KRA-26 VHF Helical Antenna (Standard Length)



KRA-41 VHF Stubby Antenna



KHS-11BL Two-wire Palm Microphone with earpiece



KHS-12BL Three-wire Lapel Microphone with earpiece



KPG-36X Full Speed USB Programming Interface Cable



KHS-14C Light Weight Single Muff Headset w/boom mic & In-line PTT



KHS-15D-OH Heavy duty noise reduction over-the- headset w/noise cancelling boom mic & PTT



KHS-15DC-BH Heavy duty behind-the-headset w/boom mic & earcup-mounted PTT with NRR of 24dB



KMC-72W Speaker Microphone



KMC-70M Speaker Microphone (with dual-sided 2-mic for superior ANR)



KBH-11 Spring Action Belt Clip (2.5")



KPG-D9R Programming Software (FPU) for NX-3210R

KPG-180AP OTAP Manager

Specifications

General	NX-3210R
Frequency Range	136-174 MHz
Max. Channels Per Radio	512 CH GID (1000 CH GID with option)
Number of Zones	128
Max. Channels per Zone	250
Channel Spacing	
Analog	25 12.5 kHz
Digital	12.5 / 6.25 kHz
Channel Step	2.5 / 3.125 / 5 / 6.25 / 7.5 kHz
Power Supply	7.5V DC $\pm$ 20%
Speaker / Mic Interface	Universal connector (14 pin)
Battery Life	(5-5-90 cycle)
KNB-L1 (2,000 mAh)	11 hours
KNB-L2 (2,600 mAh)	14 hours
KNB-L11 (3,900 mAh)	22 hours
Operating Temperature	-22°F to +140°F (-30°C to +60°C)
Storage Temperature	-40°F to +185°F (-40°C to +85°C)
Frequency Stability	$\pm$ 0.5 ppm
Dimensions Radio w/battery	(W x H x D) Projections Not Included
KNB-L1 (2,000 mAh)	2.28 x 5.49 x 1.41 in. (58.0 x 139.5 x 35.9 mm)
KNB-L2 (2,600 mAh)	2.28 x 5.49 x 1.53 in. (58.0 x 139.5 x 38.9 mm)
KNB-L11 (3,900 mAh)	2.28 x 5.49 x 1.82 in. (58.0 x 139.5 x 46.3 mm)
Weight Radio w/battery	
Radio Only	8.8 oz (250 g)
KNB-L1 (2,000 mAh)	13.4 oz (381 g)
KNB-L2 (2,600 mAh)	14.3 oz (405 g)
KNB-L11 (3,900 mAh)	16.9 oz (480 g)

Analog measurements made per TIA603. Specifications are measured according to applicable standards.
Specifications are subject to change without notice, due to advancements in technology.
Details and timing of firmware and software updates are subject to change without notice.

General	NX-3210R
Sensitivity	
NXDN 6.25 kHz Digital (3% BER)	0.20 μ V
NXDN 12.5 kHz Digital (3% BER)	0.25 μ V
Analog (12dB SINAD) @12.5 kHz	0.25 μ V
Analog (12dB SINAD) @25 kHz	0.25 μ V
Selectivity	
Analog @ 12.5kHz	65 dB
Analog @ 25kHz	72 dB
Intermodulation	70 dB
Spurious Rejection	70 dB
Audio Distortion	3 % (at 500 mW)
Audio Output Power	1 W (at 5% Distortion)
Transmitter	NX-3210R
RF Power Output	5 / 4 / 1 W
Spurious Emission	70 dB
FM Hum & Noise	
Analog @ 12.5kHz	40 dB
Analog @ 25kHz	45 dB
Audio Distortion	3%
Emission Designator	11K0F3E, 8K30F1E, 8K30F1D, 8K30F7W, 4K00F1E, 4K00F1D, 4K00F7W, 4K00F2D
Bluetooth	Bluetooth Version 4.0 Power Class 2

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MIL-STD & IP

MIL Standard	MIL 810C Methods/Procedures	MIL 810D Methods/Procedures	MIL 810E Methods/Procedures	MIL 810F Methods/Procedures	MIL 810G Methods/Procedures	MIL 810H Methods/Procedures
Low Pressure	500.1/Procedure I	500.2/Procedure I, II	500.3/Procedure I, II	500.4/Procedure I, II	500.5/Procedure I, II	500.6/Procedure I, II
High Temperature	501.1/Procedure I, II	501.2/Procedure I, II	501.3/Procedure I, II	501.4/Procedure I, II	501.5/Procedure I, II	501.7/Procedure I, II
Low Temperature	502.1/Procedure I	502.2/Procedure I, II	502.3/Procedure I, II	502.4/Procedure I, II	502.5/Procedure I, II	502.7/Procedure I, II
Temperature Shock	503.1/Procedure I	503.2/Procedure I	503.3/Procedure I	503.4/Procedure I, II	503.5/Procedure I	503.7/Procedure I
Solar Radiation	505.1/Procedure I	505.2/Procedure I	505.3/Procedure I	505.4/Procedure I	505.5/Procedure I	505.7/Procedure I
Rain	506.1/Procedure I, II	506.2/Procedure I, II	506.3/Procedure I, II	506.4/Procedure I, III	506.5/Procedure I, III	506.6/Procedure I, III
Humidity	507.1/Procedure I, II	507.2/Procedure II, III	507.3/Procedure II, III	507.4	507.5/Procedure II	507.6/Procedure II
Salt Fog	509.1/Procedure I	509.2/Procedure I	509.3/Procedure I	509.4	509.5	509.7
Dust	510.1/Procedure I	510.2/Procedure I	510.3/Procedure I	510.4/Procedure I, III	510.5/Procedure I	510.7/Procedure I
Vibration	514.2/Procedure VIII, X	514.3/Procedure I	514.4/Procedure I	514.5/Procedure I	514.6/Procedure I	514.8/Procedure I
Shock	516.2/Procedure I, II, V	516.3/Procedure I, IV	516.4/Procedure I, IV	516.5/Procedure I, IV	516.6/Procedure I, IV	516.8/Procedure I, IV, VI

International Protection Standard

Dust & Water Protection*1 IEC 60529 - IP54, IP55

ESD (Electrostatic Discharge)

ESD IEC 61000-4-2 Level 3

*1 A II interfaces must be fully sealed with appropriate covers or by genuine optional accessories.

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